

INTRODUCTION

A contactless doorbell is an advanced electronic system designed to allow visitors to ring a doorbell without physically touching any switch. In traditional doorbell systems, a person must press a button to alert the people inside the house. However, this direct

contact can lead to the spread of germs and bacteria, especially in public places and during health crises like the COVID-19. Therefore, there is a growing need for touch-free and hygienic solutions in everyday devices.

The contactless doorbell overcomes this problem by using an ultrasonic sensor, which can detect the presence of a hand or object placed near it. The sensor works by sending ultrasonic waves and measuring the distance based on the reflected signal. When a person brings their hand within a certain range, the system automatically activates a buzzer and an LED, indicating that someone is at the door. This eliminates the need for physical contact and improves safety and convenience. This project uses simple and affordable components such as an Arduino

microcontroller, ultrasonic sensor, buzzer, and LED. It is easy to build, cost-effective, and highly useful in places like homes, hospitals, offices, and public buildings. The contactless doorbell not only enhances hygiene but also represents the practical use of modern technology in daily life.

COMPONENTS AND TOOLS USED

Simulated Components (Virtual)

Breadboard

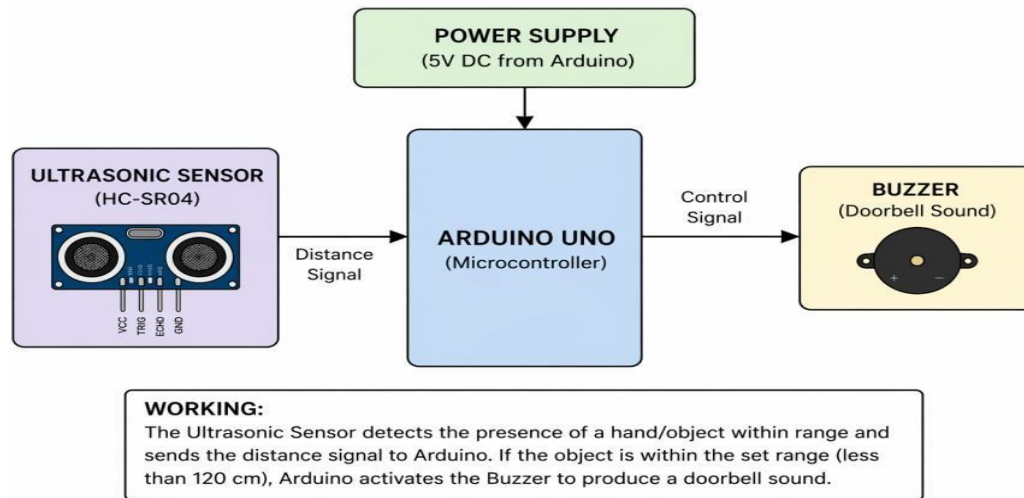
Ultrasonic sensor

Buzzer

Arduino Software Tool

Tinkercad

BLOCK DIAGRAM OF CONTACTLESS DOORBELL



SYSTEM WORKING AND OPERATION MODE

The contactless doorbell system operates based on the principle of distance measurement using an ultrasonic sensor and decision-making using a microcontroller.

When the system is powered ON, the Arduino UNO initializes all the components and starts the operation. The ultrasonic sensor continuously emits high-frequency sound waves (ultrasonic waves) into the surrounding environment through its transmitter. These waves travel in air and reflect back when they encounter an object such as a human hand.

The receiver of the ultrasonic sensor captures the reflected waves and measures the time taken for the signal to return. This time duration is sent to the Arduino, which calculates the distance using a predefined formula.

The Arduino continuously monitors this distance value. If the detected distance is within a specified threshold (for example, less than 120 cm), it interprets this as the presence of a visitor near the door.

Once a valid detection is confirmed:

- The Arduino sends a signal to activate the buzzer
- The buzzer produces a sound, functioning as a doorbell alert
- If no object is detected within the set range
- The Arduino does not trigger any output.
- The system remains in standby mode

This process repeats continuously in a loop, ensuring real-time detection and response. The system is designed to be fast, accurate, and reliable for everyday use.

RESULTS AND OUTPUT

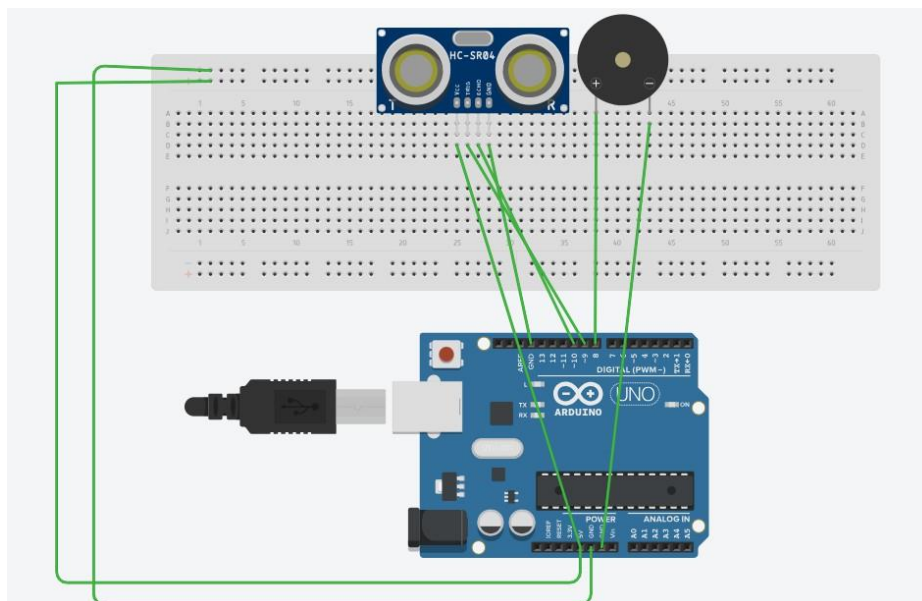
The contactless doorbell system was successfully designed and implemented using an ultrasonic sensor and Arduino UNO. The system operates effectively by detecting the presence of a hand or object without requiring any physical contact.

During testing, it was observed that when an object (such as a hand) is brought within the predefined distance range (approximately less than 120 cm), the ultrasonic sensor accurately detects it and sends the signal to the Arduino. The Arduino processes this input and activates the buzzer, producing a clear doorbell sound. When no object is present within the specified range, the system remains

inactive, and the buzzer does not produce any sound. This ensures that false triggering is minimized.

The system response was found to be quick and reliable, with minimal delay between detection and buzzer activation. The output was consistent under normal conditions, making the system suitable for real-time applications. Overall,

the project demonstrates a simple, efficient, and hygienic solution for a modern doorbell system, eliminating the need for physical contact.



CONCLUSION

The contactless doorbell system was successfully designed and implemented using an ultrasonic sensor and Arduino UNO. The system is able to detect the presence of a person without any physical contact and activates the buzzer effectively. This makes it a hygienic and user-friendly alternative to traditional doorbells.

The project is simple, low-cost, and reliable, making it suitable for practical use in homes, offices, and public places. It also demonstrates the effective use of sensors and automation in improving everyday systems. Overall, the project achieves its objective of providing a safe and contactless solution for doorbell operation.